Quantification: 682_z29
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	15
interior lacuna count	13
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 13)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1650.75	1511.5	948.377	px^2
major_axis_length_px	lacuna shape	74.0009	74.236	22.3194	px
minor_axis_length_px	lacuna shape	27.9394	29.9101	9.5919	px
aspect_ratio	lacuna shape	2.805	2.6218	0.775	unitless
eccentricity	lacuna shape	0.9144	0.9241	0.0615	unitless
solidity	lacuna shape	0.8593	0.8693	0.0551	unitless
roots_count	headline	5.4615	5	2.8465	count
ring_length_r30_px	headline	263.923	268	111.265	px
ring_length_r60_px	reported	707	732	280.787	px
owned_length_px	ownership-dependent	1882.49	1507.23	1794.82	px
edge_count	ownership-dependent	81.6154	51	83.145	count
mean_edge_length_px	ownership-dependent	27.7731	24.1355	8.7081	px
perimeter_px	normalised	186.054	202.693	70.3766	px
ring_area_r30_px2	normalised	7571.77	7711	2170.77	px^2
ring_area_r60_px2	normalised	19073	19480	4883.71	px^2
in_frame_fraction_r30	normalised	0.9952	1	0.0173	unitless
in_frame_fraction_r60	normalised	0.9839	1	0.0537	unitless
ring_density_r30	normalised	0.0348736	0.0346713	0.0108316	px^-1
ring_density_r60	normalised	0.0369335	0.036367	0.00973648	px^-1
roots_per_100px_perimeter	normalised	2.9942	2.7484	1.3714	per 100 px
ring_attached_length_r30_px	network v2	198.846	170	111.1	px
ring_attached_length_r60_px	network v2	463	356	336.899	px
ring_length_w_r30_px	network v2	294.721	292.096	121.165	px
ring_length_w_r60_px	network v2	793.152	815.855	312.916	px
sholl_crossings_r10	network v2	5.9231	6	2.7222	count
sholl_crossings_r20	network v2	9.1538	10	3.7159	count
sholl_crossings_r30	network v2	10.9231	10	3.8829	count
ring_width_mean_r30_px	width	6.9859	7.0222	0.7395	px

Per field
----

measure	value	units
analysed_area_px2	1.02273e+06	px^2
total_skeleton_length_px	33573	px
canalicular_length_density_per_px	0.0328269	px^-1
junction_count	680	count
junction_density_per_px2	0.00066489	px^-2
lacuna_count	15	count
lacunae_per_px2	1.467e-05	px^-2
skeleton_component_count	262	count

mean_component_length_px	128.141	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0370871	px^-1
field_density_without_flagged_per_px	0.0331493	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	513	px
unattached_component_count	32	count
canalicular_length_density_attached_per_px	0.0328269	px^-1
field_length_density_with_unattached_per_px	0.0333285	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5051	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	31394	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	446	49.6136	12.2658	4.0449	0.969	0.7866	-1.4445	249.987	449.359	False
4	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
5	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
6	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
7	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
8	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
9	663	61.7115	15.8655	3.8897	0.9664	0.7408	-1.1603	728.596	861.06	False
10	513	41.8409	17.3844	2.4068	0.9096	0.8607	1.3405	830.565	75.5867	False
11	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
12	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
13	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
14	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False
15								548.08	212.663	False

Network per lacuna

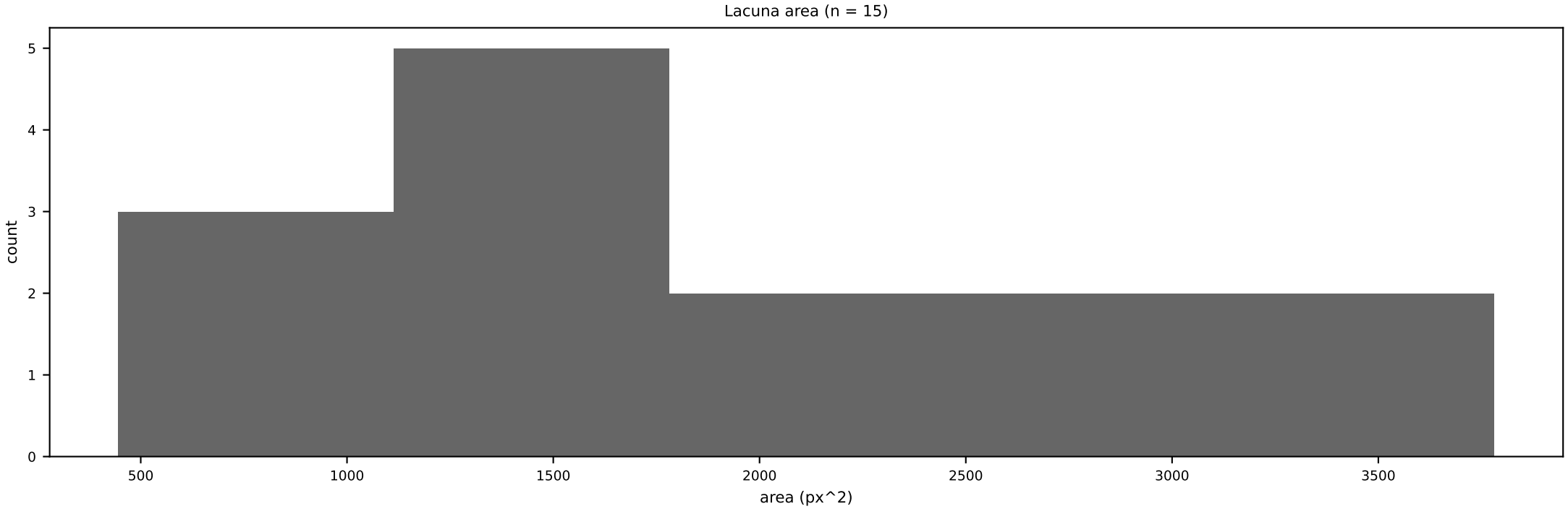
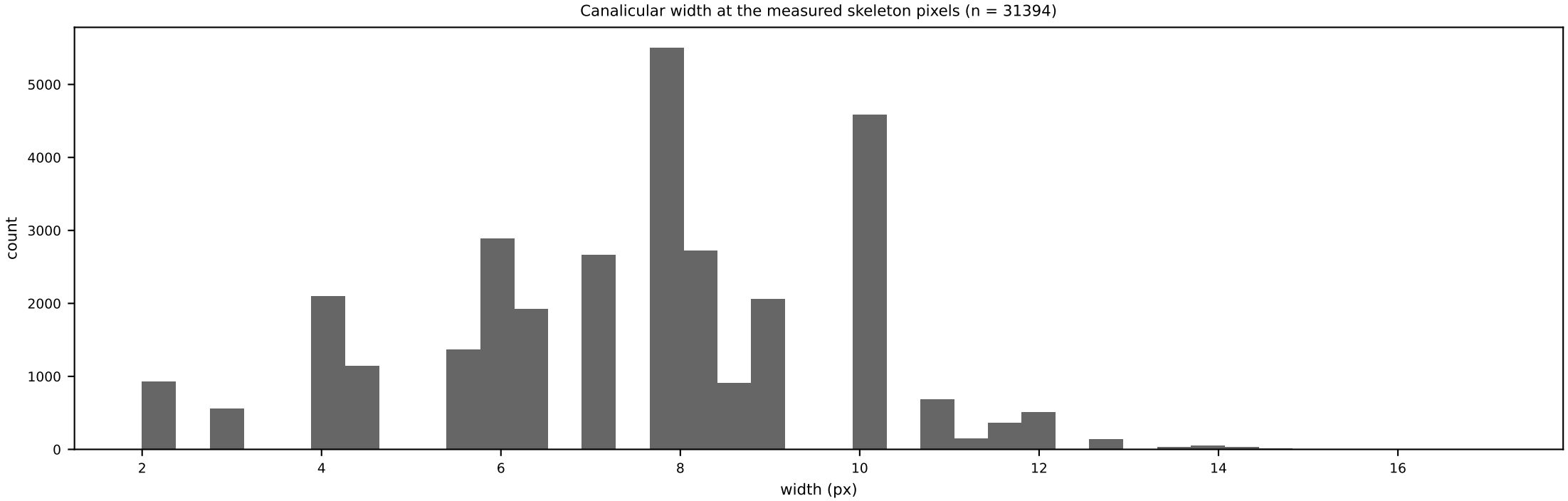
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	128	230	90.8406	8	11.3551	
2	10	463	1191	4157.8	190	21.8831	7.0028
3	5	347	918	5123.55	233	21.9895	7.4637
4	6	393	1192	4365.25	206	21.1905	7.5622
5	5	268	796	2413.55	100	24.1355	6.8365
6	7	394	858	2668.29	115	23.2025	7.3633
7	9	361	783	1341.7	55	24.3946	7.473
8	4	298	758	1507.23	51	29.5535	5.2862
9	7	176	552	2474.49	98	25.2499	6.9506

10	5	155	446	314.451	11	28.5864	6.8163
11	3	155	439	340.149	10	34.0149	7.9932
12	1	117	394	52.3553	1	52.3553	5.7831
13	7	290	732	672.6	28	24.0214	7.0222
14	10	218	381	348.22	17	20.4835	7.0835
15	1	157	534	34.3848	1	34.3848	7.6531

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0228286	0.0166437	1.1446
2	229.522	9151	23871	1	0.999	0.0505956	0.0498932	4.3569
3	120.811	6134	17911	1	1	0.0565699	0.0512534	4.1387
4	309.279	11331	27269	1	1	0.0346836	0.0437126	1.94
5	235.238	9387	21888	1	1	0.0285501	0.036367	2.1255
6	214.167	8637	19737	1	1	0.0456177	0.0434717	3.2685
7	322.836	9940	21386	0.8402	0.7724	0.0363179	0.0366127	2.7878
8	229.48	8595	22042	1	1	0.0346713	0.0343889	1.7431
9	156.853	6922	19429	1	1	0.0254262	0.0284111	4.4628
10	109.498	4245	11066	1	0.9865	0.0365135	0.0403036	4.5663
11	161.817	5491	13376	1	1	0.028228	0.03282	1.8539
12	147.196	6914	19480	1	1	0.0169222	0.0202259	0.6794
13	254.693	9728	23664	1	1	0.0298109	0.0309331	2.7484
14	202.693	7711	16371	0.9376	0.8055	0.0282713	0.0232729	4.9336
15	47.4558	4187	11845	1	1	0.037497	0.0450823	2.1072

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	20	20	145.397	253.652	1	4	4
2	377	1057	509.534	1347.5	8	16	18
3	308	902	390.078	1018.34	8	11	11
4	353	1007	437.635	1335.45	11	13	16
5	167	356	292.096	878.742	5	12	12
6	317	662	437.978	959.968	8	11	15
7	350	686	404.492	880.997	10	12	11
8	170	261	328.208	836.127	8	12	10
9	159	378	208.48	643.269	5	7	8
10	140	321	173.024	496.333	6	4	5
11	133	299	179.681	491.09	4	7	8
12	46	96	134.782	456.073	2	5	8
13	213	409	322.108	815.855	5	10	14
14	181	241	242.894	425.404	6	6	7
15	21	30	174.882	606.831	1	5	10

682_z29: distributions, pre-validation, px